

Harmonic Balance Domain Decomposition Finite Elements for Nonlinear Passive Microwave Devices Analysis

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Abstract — Full-wave analysis of the time-harmonic field originated in waveguide devices containing nonlinear materials, possibly anisotropic, is addressed. The approach combines finite elements and harmonic balance handling the nonlinearity via either Picard or relaxed iterations. To speed-up the computation, domain decomposition is exploited to restrict the iteration to a small subdomain.

Keywords — Domain decomposition method, finite element methods, harmonic analysis, nonlinear media, waveguide filters.

Running title — HB-DD-FE for Nonlinear Passive Devices Analysis

1. INTRODUCTION

Full-wave electromagnetic analysis of passive linear microwave devices can be performed via several well-established numerical methods. More complex devices, like for example those containing nonlinear, possibly anisotropic, materials are on the other hand matter of active research. Within the broad subdivision among time-domain and frequency-domain methods, the former can handle nonlinearities straightforwardly, while the second require a more complex multi-harmonic formulation. Yet, it has been proven that time-domain

analysis of waveguide devices including nonlinear materials often results to be computationally prohibitive (De Gersem et al. 2001)

As already mentioned, a steady state frequency-domain simulation requires to take into account several harmonics at the same time to model all relevant phenomena. If the frequency domain finite element method is used, then a harmonic balance finite element (HB-FE) technique (Yamada et al., 1989) or multiharmonic finite element method (Copeland & Langer, 2010) becomes mandatory. In both references, low frequency fields are considered, and in the latter, also a substructuring domain decomposition (DD) technique was also exploited. DD being a useful technique allowing to subdivide the whole domain of the problem into smaller sub-domains, each of which is coupled to the others by appropriate boundary conditions. Each domain can then be solved separately with a reduced memory and CPU time requirement within some iterative framework up to obtain the complete solution (Quarteroni & Valli, 1999; Toselli & Widlund. 2004).

Applications to higher frequencies is more recent and has been done first without resorting to harmonic balance (Guarnieri et al. 2010) and then by adding harmonic balance, but limiting the analysis to nonlinear, isotropic, permittivities (Ntibarikure et al. 2012).

In this paper materials which can be nonlinear in both permittivity and permeability will be addressed, as well as anisotropic nonlinear materials. Also the solution of the systems deriving from the DD will be handled with more efficient solvers. The capabilities of the code will be shown on few H-plane device geometries.

The paper is organized as follows: Section 2 will present the HB-FE formulation, while Section 3 will detail the DD technique. Section 4 will present some numerical results while Section 5 will draw the conclusions.

2. FORMULATION

Let's consider a generic multiport device, in general such a device must be analyzed via a full 3D numerical technique but, in many practical cases, when there is no variation of the geometry in one direction, the analysis domain can be reduced to 2D (Fig 1).

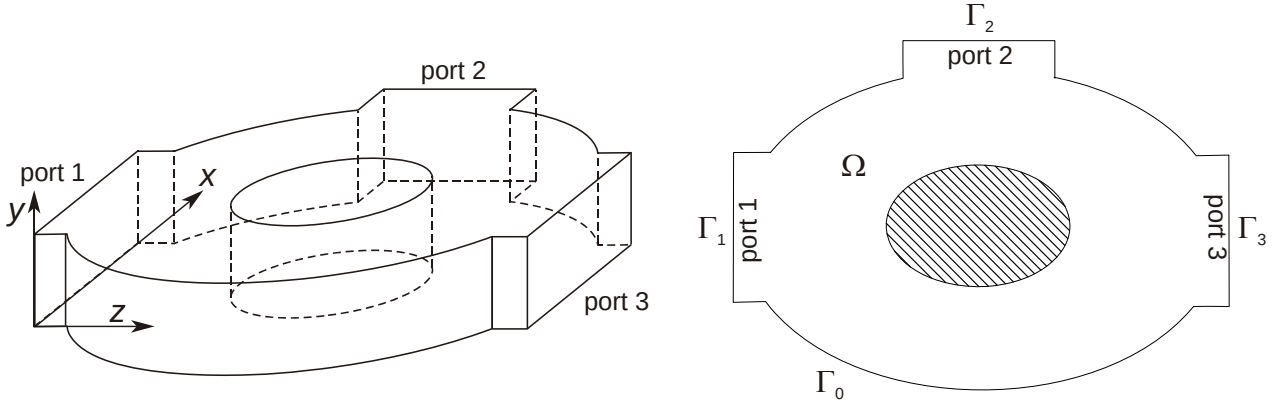


Figure 1 – Generic multiport device (left) with geometry uniform along y . and corresponding 2D domain (right) on which the full-wave analysis can be performed.

The device may contain nonlinear anisotropic dielectric characterized by:

$$\begin{aligned}\mathbf{D} &= [\varepsilon_0 \bar{\bar{\varepsilon}}_r(\mathbf{E}, \mathbf{H})] \mathbf{E} \\ \mathbf{B} &= [\mu_0 \bar{\bar{\mu}}_r(\mathbf{E}, \mathbf{H})] \mathbf{H}\end{aligned}\quad (1)$$

being ε_0 and μ_0 the permittivity and permeability of free space and $\bar{\bar{\varepsilon}}_r$ and $\bar{\bar{\mu}}_r$ the relative permittivity and permeability of the anisotropic, inhomogeneous, medium contained in the device.

The field within the device obeys to the vector Helmholtz equation:

$$\nabla \times (\bar{\bar{\mu}}_r^{-1}(\mathbf{E}(\mathbf{r}), \mathbf{r}) \nabla \times \mathbf{E}(\mathbf{r})) - k_0^2 \bar{\bar{\varepsilon}}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) = 0 \quad (2)$$

On the domain Ω , with perfect electric conductor boundary conditions on the device walls Γ_0 and a suitable modal expansion on the device ports Γ_l , for $l=1, \dots, L$, being L the number of ports (Pelosi et al. 2009, pp. 83–104).

The FE solution of (2) is well known and is obtained either via a variational formulation or a Galerkin procedure by expanding the unknown field $\mathbf{E}(\mathbf{r})$ onto a set of basis functions defined over a suitable discretization of the domain Ω in elements:

$$\mathbf{E}(\mathbf{r}) \cong \sum_{n=1}^N E_n \mathbf{a}_n(\mathbf{r}) \quad (3)$$

In any case a linear system where the unknowns are the coefficients $[E_1, E_2, \dots, E_N]$ of the expansion is obtained (Pelosi et al. 2009):

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ A_{21} & A_{22} & & A_{2N} \\ \vdots & & \ddots & \vdots \\ A_{N1} & A_{N2} & \cdots & A_{NN} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_N \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_N \end{bmatrix} \quad (4)$$

Eq. (2) is valid for harmonic field at a fixed frequency f_0 and hence for a given value of $k_0 = 2\pi f_0 \sqrt{\epsilon_0 \mu_0}$, if the medium is nonlinear and higher order harmonics are originated by it, hence there is a set of equations like (2) to be solved simultaneously, each for a different frequency, coupled by the medium.

The total electric field can be written as a Fourier series:

$$\mathbf{E}(\mathbf{r}, t) = \sum_{p=1}^{\infty} \left(\mathbf{E}_p^{(s)}(\mathbf{r}) \sin(\omega_p t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(\omega_p t) \right) \quad (5)$$

with $\omega_p = p\omega_0 = 2p\pi f_0$, to which corresponds a $k_p = pk_0$.

If the incident field contains Q frequencies f_q , $q=1, \dots, Q$ then the expansion of the electric is

$$\mathbf{E}(\mathbf{r}, t) = \sum_{\{\mathbf{p}\}} \left(\mathbf{E}_{\{\mathbf{p}\}}^{(s)}(\mathbf{r}) \sin(\omega_{\{\mathbf{p}\}} t) + \mathbf{E}_{\{\mathbf{p}\}}^{(c)}(\mathbf{r}) \cos(\omega_{\{\mathbf{p}\}} t) \right) \quad (6)$$

which has essentially the same form than (3) except that the summation is extended to all possible Q -tuples $\{\mathbf{p}\} = \{p_1, p_2, \dots, p_Q\}$ with $p_q \in \mathbb{Z}$ and $\omega_{\{\mathbf{p}\}} = p_q \omega_q$ with $\omega_q = 2\pi f_q$.

Since also material parameters can be express in terms of a Fourier series holding both for the case (4) and for the case (3) if $Q=1$ and the Q -tuple reverts to a single integer (Yamada et al., 1989; Copeland & Langer, 2010):

$$[\bar{\bar{\epsilon}}_r(\mathbf{E}, \mathbf{H}), \bar{\bar{\mu}}_r(\mathbf{E}, \mathbf{H})] = \sum_{\{\mathbf{g}\}} \left([\bar{\bar{\epsilon}}_{r_{\{\mathbf{g}\}}}^{(s)}, \bar{\bar{\mu}}_{r_{\{\mathbf{g}\}}}^{(s)}] \sin(\omega_{\{\mathbf{g}\}} t) + [\bar{\bar{\epsilon}}_{r_{\{\mathbf{g}\}}}^{(c)}, \bar{\bar{\mu}}_{r_{\{\mathbf{g}\}}}^{(c)}] \cos(\omega_{\{\mathbf{g}\}} t) \right) \quad (7)$$

with

$$\begin{aligned} [\bar{\bar{\epsilon}}_{r_{\{\mathbf{g}\}}}^{(s)}, \bar{\bar{\mu}}_{r_{\{\mathbf{g}\}}}^{(s)}] &= \frac{\omega}{\pi} \int_0^{2\pi} [\bar{\bar{\epsilon}}_r(\mathbf{E}, \mathbf{H}), \bar{\bar{\mu}}_r(\mathbf{E}, \mathbf{H})] \sin(\omega_{\{\mathbf{g}\}} t) dt \\ [\bar{\bar{\epsilon}}_{r_{\{\mathbf{g}\}}}^{(c)}, \bar{\bar{\mu}}_{r_{\{\mathbf{g}\}}}^{(c)}] &= \frac{\omega}{\pi} \int_0^{2\pi} [\bar{\bar{\epsilon}}_r(\mathbf{E}, \mathbf{H}), \bar{\bar{\mu}}_r(\mathbf{E}, \mathbf{H})] \cos(\omega_{\{\mathbf{g}\}} t) dt \end{aligned} \quad (8)$$

The infinite summations in (6) and (7) are truncated to a finite number of terms, respectively P and G to allow for numerical computation. A possible truncation criterion, as illustrated in Section 4, is to limit the harmonic expansions such that the impinging power at fundamental frequencies does not vary more than a prescribed value while the order is increased (Ntibarikure et al. 2012). In this framework (2) must be solved

for all harmonic fields $\mathbf{E}_{\{p\}}^{(s,c)}$ in expansion (6), hence expansion (3) must be applied to all $\mathbf{E}_{\{p\}}^{(s,c)}$ and, the linear system (4) stays formally the same except that each of the entry A_{mn} of matrix $[\mathbf{A}]$ becomes itself a $2P \times 2P$ matrix while each entry E_n of vector $[\mathbf{E}]$ becomes a $2P$ vector, relative to the P sine and P cosine coefficients in the expansion (6), the same holds true for the known vector $[\mathbf{B}]$ terms :

$$A_{mn} = \begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \dots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \dots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \dots & a_{mn}^{(PcPc)} \end{bmatrix}; \quad E_n = \begin{bmatrix} e_n^{(1s)} \\ e_n^{(1c)} \\ \vdots \\ e_n^{(Ps)} \end{bmatrix} \quad B_m = \begin{bmatrix} b_m^{(1s)} \\ b_m^{(1c)} \\ \vdots \\ b_m^{(Pc)} \end{bmatrix} \quad (9)$$

in (9) indices p and q in $a_{mn}^{(p[s,c]q[s,c])}$ are relative to an appropriate ordered indexing of all the possible Q -tuples $\{\mathbf{p}\}$. In the case of a single incident frequency then p is the order of the harmonic.

Since $a_{mn}^{(p[s,c]q[s,c])}$, and hence A_{mn} depends on the material properties (1) through (2), and hence on the electric and magnetic fields, it is evident that entries of matrix $[\mathbf{A}]$ depend on the solution $[\mathbf{E}]$, hence a direct solution is not possible.

An iterative solution is hence necessary, hypothesizing a null first guess $[\mathbf{E}]^{(0)}$ for the unknown electric field to compute a first system matrix $[\mathbf{A}([\mathbf{E}]^{(0)})]$ and right hand side $[\mathbf{B}([\mathbf{E}]^{(0)})]$. From these a new value of the unknown vector is obtained solving $[\mathbf{A}_{([\mathbf{E}]^{(0)})}][\mathbf{E}]^{(1)} = [\mathbf{B}_{([\mathbf{E}]^{(0)})}]$. From the new value $[\mathbf{E}]^{(1)}$ a new system can be computed.

Rather than the simple Picard iteration detailed above, a relaxed iteration often proves to have better convergence properties. In the relaxed iteration the field used to compute the new linear system is a weighted sum of the two previous iteration step results:

$$[\mathbf{A}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}][\mathbf{E}]^{(i)} = [\mathbf{B}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}], \quad i \geq 2 \quad (10)$$

with $\gamma \in (0,1]$. $\gamma=1$ gives the standard Picard iteration (Guarnieri et al., 2010), while low values damps the oscillations which may arise with highly nonlinear materials or high intensity impressed fields at the cost of a slower convergence (Silvester & Ferrari, 1997, pp. 187–190). The process is repeated until the relative error between the updated solution and the previous one, in the sense of the Euclidean norm, is less than a prescribed value (Guarnieri et al. 2010, Ntibarikure et al. 2012):

$$\frac{\|[\mathbf{E}]^{(i)} - [\mathbf{E}]^{(i-1)}\|}{\|[\mathbf{E}]^{(i)}\|} \leq \tau \quad (11)$$

The order of the resulting HB-FE systems (10) is hence $2PN$, and this number can become very high, especially if more than a frequency is present in the impinging signal and intermodulation products are generated. Since, furthermore, the solutions have to be iterated building at each step a new linearized system, computing time can become too high. To enhance the computational efficiency, a DD approach can be exploited, as detailed in the following section.

3. DOMAIN DECOMPOSITION

In the proposed DD approach the original problem domain Ω (Fig. 1) is decomposed in several smaller subdomains Ω_i so that (Fig. 2):

$$\Omega = \bigcup_{i=1}^N \Omega_i; \quad \Omega_i \cap \Omega_j = \emptyset \quad \forall i \neq j \quad (12)$$

The unknowns of the original FE problem (4) can be renumbered on a subdomain by subdomain basis so as to group together all unknowns internal to each subdomain, while keeping together all the unknowns relative to all boundaries $\Gamma_0, \Gamma_I, \Gamma_{i,j}$.

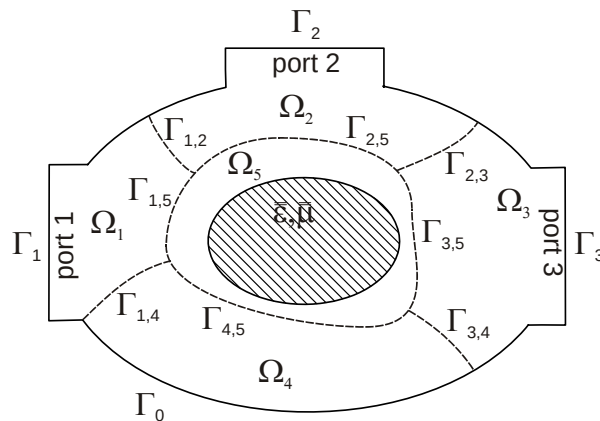


Figure 2 – Generic multiport device (left) with geometry uniform along y . and corresponding 2D domain (right) on which the full-wave analysis can be performed.

By naming $[\mathbf{E}]_i$ the vector of the unknowns internal to Ω_i , $[\mathbf{E}]_d = [\mathbf{E}]_1, [\mathbf{E}]_2, \dots, [\mathbf{E}]_N$ the vector of all unknowns internal to all subdomain, and by naming $[\mathbf{E}]_c$ the vector of the unknowns on all boundaries, and by performing a similar operation on the right hand side term, (4) can be rewritten as

$$[\mathbf{A}][\mathbf{E}] = [\mathbf{B}] \Leftrightarrow \begin{bmatrix} [\mathbf{M}] & [\mathbf{P}] \\ [\mathbf{Q}] & [\mathbf{C}] \end{bmatrix} \begin{bmatrix} [\mathbf{E}]_d \\ [\mathbf{E}]_c \end{bmatrix} = \begin{bmatrix} [\mathbf{B}]_d \\ [\mathbf{B}]_c \end{bmatrix} \Leftrightarrow \left[\begin{array}{cccc|c} [\mathbf{M}]_1 & 0 & \cdots & 0 & [\mathbf{P}]_1 \\ 0 & [\mathbf{M}]_2 & \cdots & 0 & [\mathbf{P}]_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & [\mathbf{M}]_N & [\mathbf{P}]_N \\ \hline [\mathbf{Q}]_1 & [\mathbf{Q}]_2 & \cdots & [\mathbf{Q}]_N & [\mathbf{C}] \end{array} \right] \begin{bmatrix} [\mathbf{E}]_1 \\ [\mathbf{E}]_2 \\ \vdots \\ [\mathbf{E}]_N \\ [\mathbf{E}]_c \end{bmatrix} = \begin{bmatrix} [\mathbf{B}]_1 \\ [\mathbf{B}]_2 \\ \vdots \\ [\mathbf{B}]_N \\ [\mathbf{B}]_c \end{bmatrix} \quad (13)$$

where the block structure of matrix $[\mathbf{A}]$ is evident. The form (13) of course holds also for the multiharmonic case if each entry in c is itself a matrix (9).

If the domains are arranged so that the nonlinear anisotropic material is fully contained in the last domain Ω_N then (13) can be rewritten with its explicit dependence on the iteration (10) as

$$\left[\begin{array}{cccc|c} [\mathbf{M}]_1 & 0 & \cdots & 0 & [\mathbf{P}]_1 \\ 0 & [\mathbf{M}]_2 & & \vdots & [\mathbf{P}]_2 \\ \vdots & & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & [\mathbf{M}_{(\gamma[\mathbf{E}]^{i-1} + (1-\gamma)[\mathbf{E}]^{i-2})}]_N & [\mathbf{P}_{(\gamma[\mathbf{E}]^{i-1} + (1-\gamma)[\mathbf{E}]^{i-2})}]_N \\ \hline [\mathbf{Q}]_1 & [\mathbf{Q}]_2 & \cdots & [\mathbf{Q}_{(\gamma[\mathbf{E}]^{i-1} + (1-\gamma)[\mathbf{E}]^{i-2})}]_N & [\mathbf{C}_{(\gamma[\mathbf{E}]^{i-1} + (1-\gamma)[\mathbf{E}]^{i-2})}] \end{array} \right] \begin{bmatrix} [\mathbf{E}]_1^{(i)} \\ [\mathbf{E}]_2^{(i)} \\ \vdots \\ [\mathbf{E}]_N^{(i)} \\ [\mathbf{E}]_c^{(i)} \end{bmatrix} = \begin{bmatrix} [\mathbf{B}]_1 \\ [\mathbf{B}]_2 \\ \vdots \\ [\mathbf{B}_{(\gamma[\mathbf{E}]^{i-1} + (1-\gamma)[\mathbf{E}]^{i-2})}]_N \\ [\mathbf{B}_{(\gamma[\mathbf{E}]^{i-1} + (1-\gamma)[\mathbf{E}]^{i-2})}]_c \end{bmatrix} \quad (14)$$

In (14) it is evident how the larger part of $[\mathbf{A}]$ does not change during the iteration, updating being limited to the four blocks in the lower right corner.

By using the Schur complement concept (Saad, 2000), the boundary unknowns of (14) and hence the generalized scattering matrix of the device can be retrieved with the same accuracy of the full domain solution (Guarnieri et al., 2010; Ntibarikure et al., 2012).

$$[\mathbf{S}]^{(i-1)}[\mathbf{E}]_c^{(i)} = [\mathbf{D}]^{(i-1)} \quad (15)$$

with

$$[\mathbf{S}]^{(i-1)} = [\mathbf{C}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}] - \left\{ \sum_{n=1}^{N-1} [\mathbf{Q}]_n [\mathbf{M}]_n^{-1} [\mathbf{P}]_n \right\} - [\mathbf{Q}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}]_N [\mathbf{M}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}]_N^{-1} [\mathbf{P}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}]_N \quad (16)$$

and

$$\begin{aligned}
[\mathbf{D}]^{(i-1)} = & [\mathbf{B}_{(\gamma[E]^{(i-1)} + (1-\gamma)[E]^{(i-2)})}] - \left\{ \sum_{n=1}^{N-1} [\mathbf{Q}]_n [\mathbf{M}]_n^{-1} [\mathbf{P}]_n \right\} - \\
& [\mathbf{Q}_{(\gamma[E]^{(i-1)} + (1-\gamma)[E]^{(i-2)})}]_N [\mathbf{M}_{(\gamma[E]^{(i-1)} + (1-\gamma)[E]^{(i-2)})}]_N^{-1} [\mathbf{B}_{(\gamma[E]^{(i-1)} + (1-\gamma)[E]^{(i-2)})}]_N
\end{aligned} \tag{17}$$

In (16) and (17) the terms in braces, containing computationally intensive inversions, need to be computed only once before starting the iteration, thus, in order to solve the HBDDFE system, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_N and Γ , and hence the first and last term in the right hand side of (16) and (17) are to be updated.

The efficiency of this DD technique relies on the restriction of matrices updating to sufficiently small regions along the nonlinear iterative solution, noticeably improving long nonlinear loops. The Schur complement matrices related to linear materials domains, remaining constant all over the iterative solution loop, have to be assembled only once. The computation of the Schur complements with conventional techniques leads to a small but dense matrix (Toselli & Widlund., 2004; Guarnieri et al. 2010), which is known leading to high computational complexity for large nonlinear subdomains. One way to reduce the complexity of the Schur complement based domain decomposition could be the use of an iterative solver, with proper preconditioner, for the boundary unknowns solution (Gaidamour et al., 2008).

4. NUMERICAL RESULTS

To assess the validity of the proposed approach, the test case of a H-plane circulator in rectangular waveguide constituted by a Y-junction with a magnetized ferrite post is presented (Koshiba & Suzuki, 1986). The circulator comprises three WR90 (cross- section dimensions $a = 22.86\text{mm}$; $b = 10.16\text{mm}$) waveguide sections of length $l = 33.4\text{mm}$; joined forming a 120° angle one with each other in a Y shape and a magnetized ferrite post of height b , hence spanning the whole device height. The post is an equilateral triangular prism of base side $d = 7.80\text{mm}$, placed at the center of the junction as sketched in Fig. 3 and it is magnetized along z .

While the most of the device, Ω_1 , is in air ($\epsilon_r = \mu_r = 1$), the post, Ω_2 , is made of a nonlinear magnetized ferrite. In (Koshiba & Suzuki, 1986) and in many other papers, the ferrite is considered linear and characterized by:

$$\bar{\bar{\mu}}_r(\mathbf{r}) = \begin{bmatrix} \mu_r & j\kappa_r & 0 \\ -j\kappa_r & \mu_r & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad \bar{\bar{\varepsilon}}_r(\mathbf{r}) = \varepsilon_r \quad (18)$$

with, as in the standard linear approach in which only the static (DC) external impressed magnetic field is considered and the harmonic (AC) signal neglected:

$$\mu_r = 1 + \frac{(\omega_0 + j\omega\alpha)\omega_m}{(\omega_0 + j\omega\alpha)^2 - \omega^2}; \quad \kappa_r = \frac{\omega\omega_m}{(\omega_0 + j\omega\alpha)^2 - \omega^2} \quad (19)$$

being $\omega_0 = \gamma H_i$, $\omega_m = \gamma M_s$, $\alpha = \gamma \Delta H / 2\omega$; and being H_i the external DC magnetic impressed field, uniform along the out of plane direction, M_s the saturation magnetization, ΔH the resonance linewidth and γ the gyromagnetic ratio. In the following the values considered will be: $\gamma = 1.76 \cdot 10^7 \text{ C kg}^{-1}$; $H_i = 200 \text{ G}$; $M_s = 1317 \text{ Oe}$; $\Delta H = 135 \text{ Oe} \cdot \text{s}$ and $\varepsilon_r = 11.7$ as reported in (Koshiba et al., 1986)

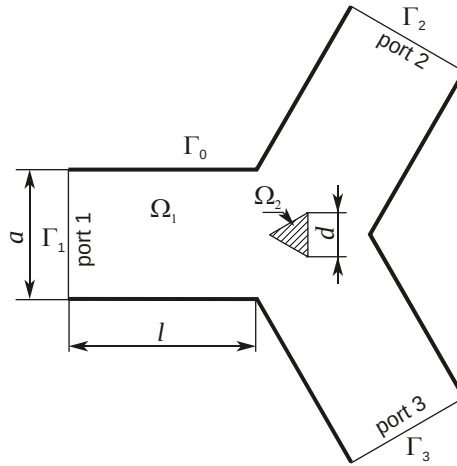


Figure 3 – Geometry layout of the ferrite circulator. The two domains Ω_1 and Ω_2 for the DD approach are also shown.

Eq. (18) still holds for non-linear ferrites if the AC magnetic field is not neglected with respect to the external DC magnetic field. In this case $\mathbf{H} = H_i \hat{z} + \mathbf{h}_{AC}$, $\mathbf{M} = M_s \hat{z} + \mathbf{m}_{AC}$ and, upon derivation from the macroscopic equation of motion for large signals and assumption of the z-directed components of the AC field to be neglectable, (19) are replaced by:

$$\mu_r = 1 + \frac{(\omega_0 + j\omega\alpha)\omega_m}{(\omega_0 + j\omega\alpha)^2 + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2}; \quad \kappa_r = \frac{\omega\omega_m}{(\omega_0 + j\omega\alpha)^2 + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2} \quad (20)$$

being h_x and h_y the two components, on the H-plane, of the AC magnetic field.

The analysis is first conducted assuming small signals impinging on Port 1 (Γ_1), hence with linear permeability tensor. The resulting spectral response is given in Fig. 4, and it matches the one reported by (Koshiba et al. 1986). Figure 5 reports the corresponding fieldmap for a 9GHz signal.

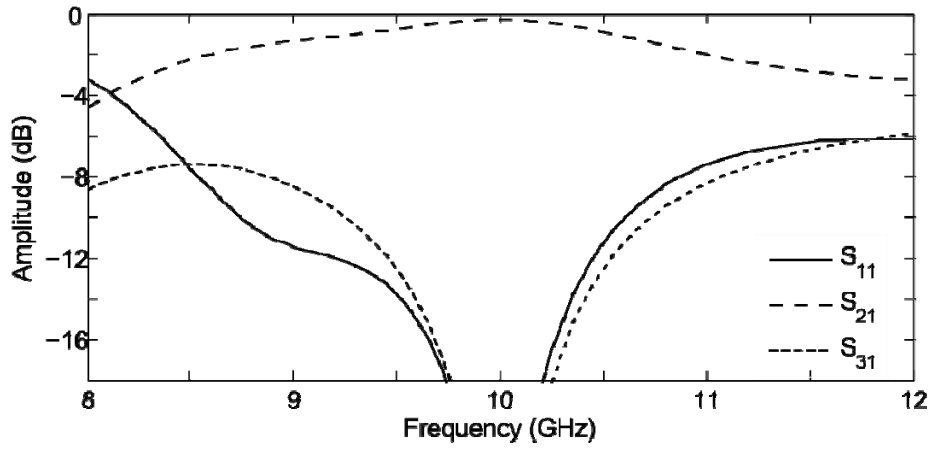


Figure 4. Circulator's small signal frequency response.

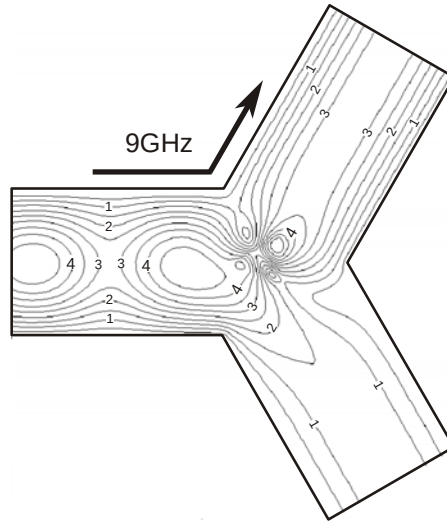


Figure 5. Electric field (kV/m) distribution at 9 GHz for 1 W impinging on Γ_1 .

To assess nonlinearity of the modeled device, the circulator is analyzed assuming a strong interfering signal impinging on the isolated port (Γ_3) at $f_i = 10$ GHz, 10 dB higher than the signal at Port 1, being the signal

frequency $f_s \in [8, 12]$ GHz, engendering undesired intermodulation products (IMPs) at the coupled port (Γ_2). The convergence of the harmonic orders (P) is analyzed upon increasing the order progressively and monitoring the power kept by in-band IMPs signals ($2f_i - f_s$ and $2f_s - f_i$). The order of G is chosen such that $G\omega_g = \max(\omega_{(p)})$, $\omega_g = \gcd(\omega_i, \omega_s)$. For this test, the signal power impinging at $f_s = 9$ GHz has been assumed of 150 W, and the interferer one of 1.5 kW at $f_i = 10$ GHz. An iteration loop stop criterion $\tau = 10^{-9}$ has been assumed. Results are shown in Table 1, while Fig. 6 reports the fieldmaps at the signal and interferent frequencies, as well as the in-band interferent.

It is worth noticing that as the harmonic order grows, higher order TE modes at ports boundaries become propagating, and hence 10 modes (Pelosi et al. 2009) have been retained to accurately compute the power balance.

Table 1. Harmonic power at third order intermodulation products as P is increased.

	Harmonic order considered (cumulative from left to right)			
Harmonic order	Only $2f_1-f_2, 2f_2-f_1$	$2f_1+f_2, 2f_2+f_1, 3f_1, 3f_2$ (all 3 rd order)	$f_1+f_2, f_1-f_2 , 2f_1, 2f_2$ (all 2 nd order)	$3f_1-2f_2, 3f_2-2f_1, 3f_1+2f_2, 3f_2+2f_1$
Power at $2f_1-f_2$ (W)	$4.6019 \cdot 10^{-3}$	$4.2246 \cdot 10^{-3}$	$4.2246 \cdot 10^{-3}$	$4.2186 \cdot 10^{-3}$
Power at $2f_2-f_1$ (W)	$2.0990 \cdot 10^{-1}$	$2.1951 \cdot 10^{-1}$	$2.1951 \cdot 10^{-1}$	$2.1991 \cdot 10^{-1}$

Even if the exact (continuous) solution cannot be achieved, Table 1 shows that the computation of in-band third order products, considering up to all the third order products, leads to a maximum error of less than 0.2%. In this case, Fig. 7 and 8 shows the resulting power amplitudes as a function of frequency for the two in-band third-order IMPs for different signal and interferent power levels.

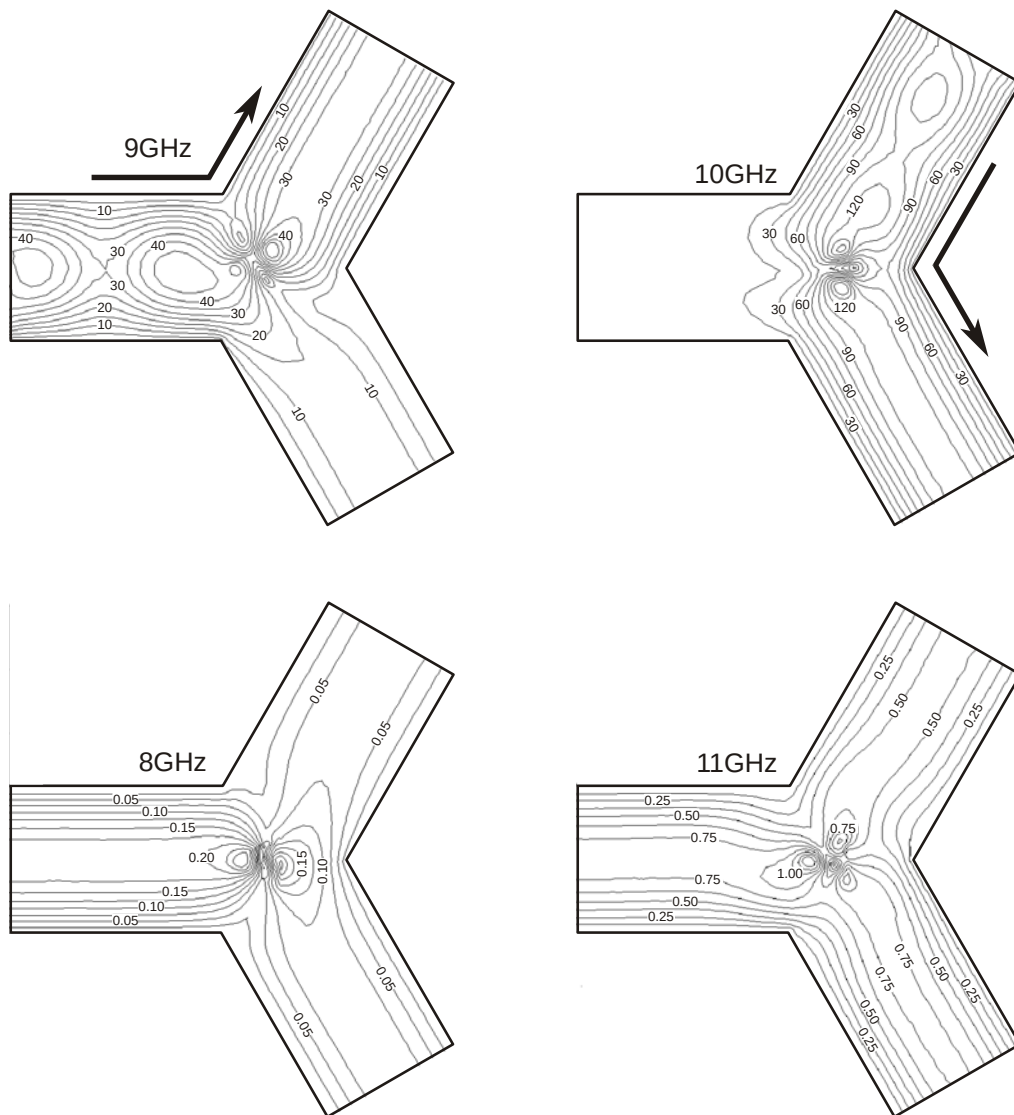


Figure 6. Electric field (kV/m) distribution for the third order intermodulation problem, at 9 GHz, 10 GHz, and at in-band IMPs frequencies 8 and 11 GHz.

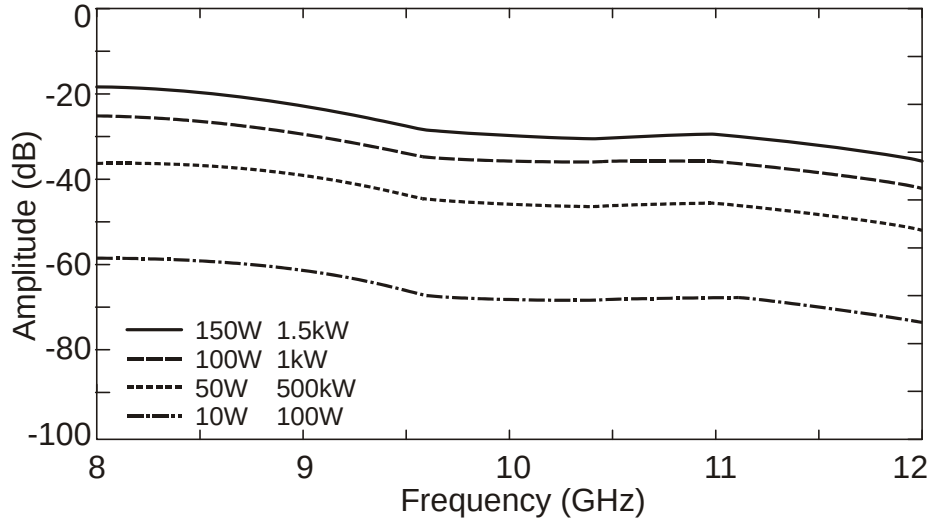


Figure 7. Intermodulation product power at $2f_s - f_i$

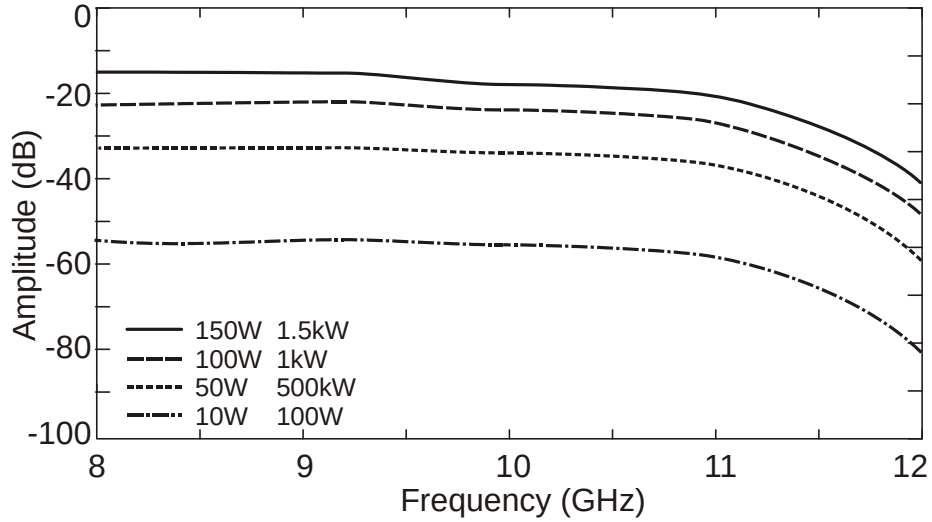


Figure 8. Intermodulation product power at $2f_i - f_s$

The power delivered to IMPs is in good agreement with those reported in (How et al. 1997).

The use of the DD scheme, which provides the same solution of the standard full system iteration scheme, allows to speed up the computations. The acceleration of the DD solver, defined as the ratio between the standard scheme times (assembly and solve for whole domain at each iteration) and the DD times (assembly and solve for whole domain only at first iteration, then assembly and solve for nonlinear subdomains only), is reported in Fig. 9. Sparse direct solvers of a Matlab® implementation for both standard full and DD schemes have been employed to perform the computations.

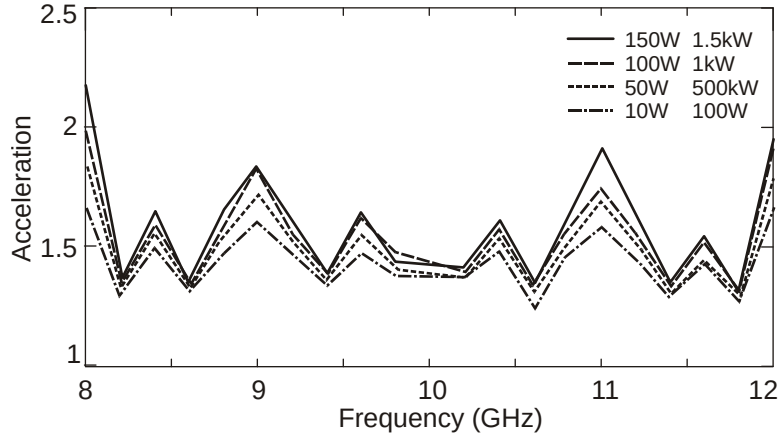


Figure 9. Acceleration of the DD scheme vs conventional scheme. Linear elements have been considered, leading to 700 degrees of freedom and 8100 non-zero entries of $[A]$.

All over the frequency sweep, the FFT-based algorithm employed to compute nonlinear materials testing has required noticeably different amount of times, bound to differences in the number of iterations required, leading to different values of acceleration for the chosen frequency points. The time step required to accurately compute the testing integrals strongly depend on the difference between signal and interferer frequencies. However, 45% to 60% average speed-up have been noticed within the signal bandwidth.

Further test are performed to assess the efficiency of the method, both for 2 and 5 subdomains (only 1 with nonlinear ferrite), while increasing the number of degrees of freedom (hence the non-zero entries of the system matrix $[A]$) by increasing the elements order from first to fourth (Fig 10 and 11).

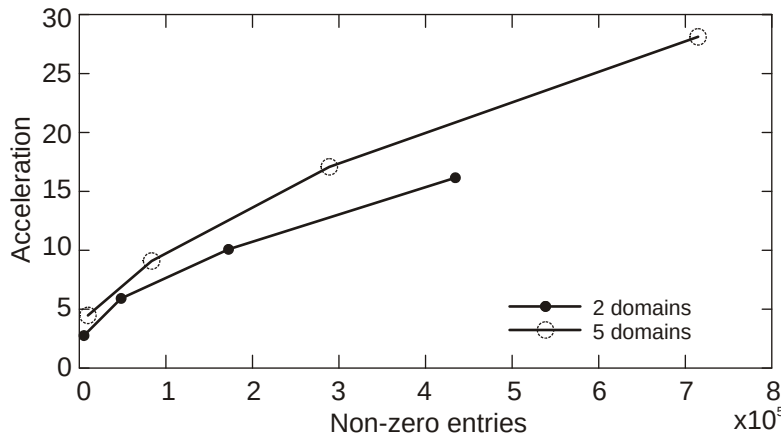


Figure 10. Acceleration of the DD scheme vs standard full scheme as a function of non-zero entries of $[A]$.

The higher is the number of linear subdomains, the smaller are the submatrices of the linear subdomains, leading to improved acceleration. Furthermore, acceleration grows logarithmically with the polynomial order, that is with the number of non-zero entries of the system matrix.

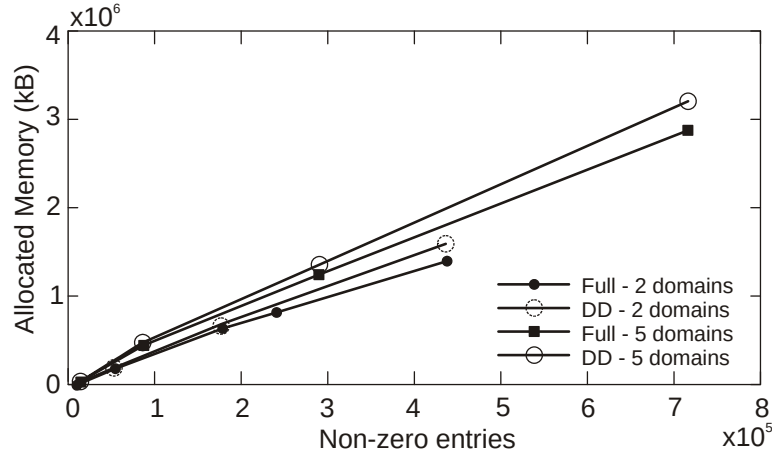


Figure 11. Allocated memory (stored and dynamically freed) as a function of non-zero entries of $[A]$.

Increasing the number of subdomains generally leads to higher allocated memory. This is mainly due to an increase of geometrical and degrees of freedom reordering information, which is kept in both standard full and DD schemes. DD schemes require more memory to store the Schur complement matrices and solve for related unknowns

5. CONCLUSIONS

A finite element approach to the analysis of nonlinear passive microwave devices has been presented. The paper presents the Harmonic Balance as a valid methodology for the calculation of harmonic couplings occurring within nonlinear materials characteristics at microwave frequencies. A Schur-complement based domain decomposition technique has been applied to speed-up the nonlinear iterative solution.

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